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Sen2Chain: An Open-Source Toolbox for Processing Sentinel-2 Satellite Images and Producing Time-Series of Spectral Indices

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Summary

Satellite data for Earth observation has become essential for monitoring natural and human processes. The increasing volume of publicly accessible remote sensing data brings many opportunities, but also operational challenges, that continue to limit its use across a broad range of users. Processing chains promote the use of this data by automating the tedious work of repetitively processing large datasets. They are important for fully leveraging the repetitive nature of the imagery by creating time series generated from past satellite missions, and enabling near-real-time automated updates as new images are acquired. Sen2Chain allows for heavy processing of Sentinel-2 (S2) satellite images by offering functions dedicated to the management of large databases, notably through the use of parallelization and automation. Its operation is quite straightforward, with object-oriented programming, for easy manipulation of data and functions. It was developed for the needs of scientific research and operational programs aimed at producing short-term environmental indicators for early warning systems, which required large-scale, high-resolution spatial environmental data (10 meters) over long periods and at high temporal frequencies (every 5 days). Its free and open-source nature enhances its accessibility, and broadens its potential range of use.

Statement of need

The high availability of large open access satellite image datasets, whose spatial and spectral characteristics are constantly improving, currently makes it possible to consider operational analyses covering vast areas. Time series of environmental indices from Earth observation (EO) data (e.g., vegetation, water or humidity indices) are essential for monitoring and predicting land surface dynamics. Assessing the changes of individual pixel values over time through various spectral indices is a common approach in remote sensing research (Hislop et al., 2018). These time-series capture seasonal environmental fluctuations and local dynamics that may be linked to human practices such as deforestation (Gao et al., 2020; Laag & Pilario, 2026), agriculture (?), water quality (Pahlevan et al., 2022), fire monitoring and mapping (Huang et al., 2016; Verhegghen et al., 2016), but also natural trends or exceptional weather events such as floods or cyclones, which are heavily influenced by climate change (Sanyal & Lu, 2004). For these applications, since 2015, the exploitation of time series of EO data, using the free S2 optical satellite constellation of the European Space Agency's (ESA) Copernicus program, has marked an important step, bringing a significant gain compared to previous satellites in terms of spatial and temporal resolutions (up to 10 meters and 5 days respectively). These characteristics have already proven their worth in a wide range of fields, such as the mapping of land-cover (Phiri et al., 2020), the impact of natural disasters (Mouquet et al., 2020) or the mapping of habitats (Poursanidis et al., 2019). Users of S2 images benefit from pre-processed data (L2A level), orthorectified (taking into account geometric distortions related to the terrain and the sensor) and atmospherically corrected, directly usable for their applications. They also have free access to powerful ESA tools designed for advanced analysis, such as SNAP and Sen2Cor. Combined with the ease of access to S2 data through a web portal and the Copernicus Data Space Ecosystem (CDSE) API, this has opened up very interesting prospects for scientific research and the development of operational products. Such a context has encouraged the development of tools to process complete time series of images, particularly for the serial production of environmental indices. This article introduces Sen2Chain, a processing chain designed to download, process, and manage large volumes of S2 data covering broad temporal and spatial extents. Sen2Chain was designed to enable users of all programming backgrounds to generate and manage large amounts of Earth observation data, by automating and parallelizing the processing of S2 images, and producing consistent time series of radiometric indices over a given period and for a geographic area.

State of the field

Sen2Chain is developed in Python and runs similarly to existing dedicated S2 image processing applications, such as sen2r (Ranghetti et al., 2020) whose development based on the R language has stopped since 2023, complementing it with more specific features. For instance, it integrates an advanced image download tool, supports easy management of an image and derivative database, allows automatic extraction of statistics in regions of interest (ROI), automates the processing of vast geographical areas and time periods. The tool also stands out from Google Earth Engine (GEE), the Google's online processing platform that also allows processing of large volumes of data from Earth observation satellites. GEE offers many advantages and is increasingly used worldwide for scientific research (Velastegui-Montoya et al., 2023), especially by users with limited resources (Johary et al., 2023; Vijayakumar et al., 2024). But it is not without constraints either and includes limitations such as limited free storage space, restrictions on freely available processing tools, limitations in terms of data sharing with other users and rules of use, intellectual property rights and confidentiality governed by Google (Amani et al., 2020; Padarian et al., 2015). Furthermore, evolving geopolitical and economic contexts may influence the policies of platforms such as Google Earth Engine (GEE), a service operated by a private U.S.-based company, potentially affecting future accessibility and use. Projects, organizations, or countries that have invested in the platform for large-scale operational developments could find themselves blocked and their needs unmet if their access or quotas are no longer guaranteed. With opensource tools like Sen2Chain, users retain control over their entire processing chains, from the early stages of development to the final results.

Software description

Sen2Chain is entirely developed in Python, using free and open-access software libraries developed by the user community to efficiently perform various functions. Along with R, Python is one of the most widely used open source languages for scientific and geospatial applications (Canty, 2025; Kaya et al., 2019). Since the 2000s, many popular high-performance libraries have been developed in Python for raster and vector GIS data management, image processing, and remote sensing. The most common and widely used libraries include GDAL, Rasterio, Pandas, Numpy, SciPy, and MatPlotLib. Packages and tools more specific to the exploitation of S2 imagery were also used, such as EODAG for data download, Sen2Cor for atmospheric corrections, and OmniCloudMask for cloud detection. Sen2Chain was designed to handle the large volumes of data involved in processing S2 images with high spatial resolution, high temporal frequency, and global coverage. The database, stored locally on the user's workstation or on a computing server, can easily contain tens of thousands of images representing tens of terabytes of data. The basic processing unit of Sen2Chain is the S2 tile, a 110 x 110 km^2 orthoimage with a WGS84/UTM geographic projection based on the Military Grid Reference System (MGRS) with global coverage. Each tile has a unique identifier to which all data is associated, from the raw images (L1C level) to the final spectral indices produced.

Key features

Sen2Chain allows downloading S2 images from different online catalogs using the EODAG library (Figure 1). Users can choose either to download the raw L1C images to produce their own L2A images, according to their needs and parameters using ESA's Sen2Cor tool for preprocessing, or to download the L2A directly from online catalogues, thus reducing resources and processing time. After querying the online catalogs, downloads are performed by geographic zones, using the identifiers of the tiles of interest, and user-defined maximum cloud cover and temporal extent. The L2A images are then used to generate cloud masks and spectral indices (e.g., brightness, vegetation, and humidity indices), whose values can be extracted from areas of interest for the needs of different applications. Sen2Chain can produce different versions of cloud masks, either directly using cloud-related classes derived from the L2A Scene Classification product, or using the open-source OmniCloudMask library, which employs sensor-independent deep learning models and offers excellent cloud detection performance for Sentinel-2 data (Wright et al., 2025).

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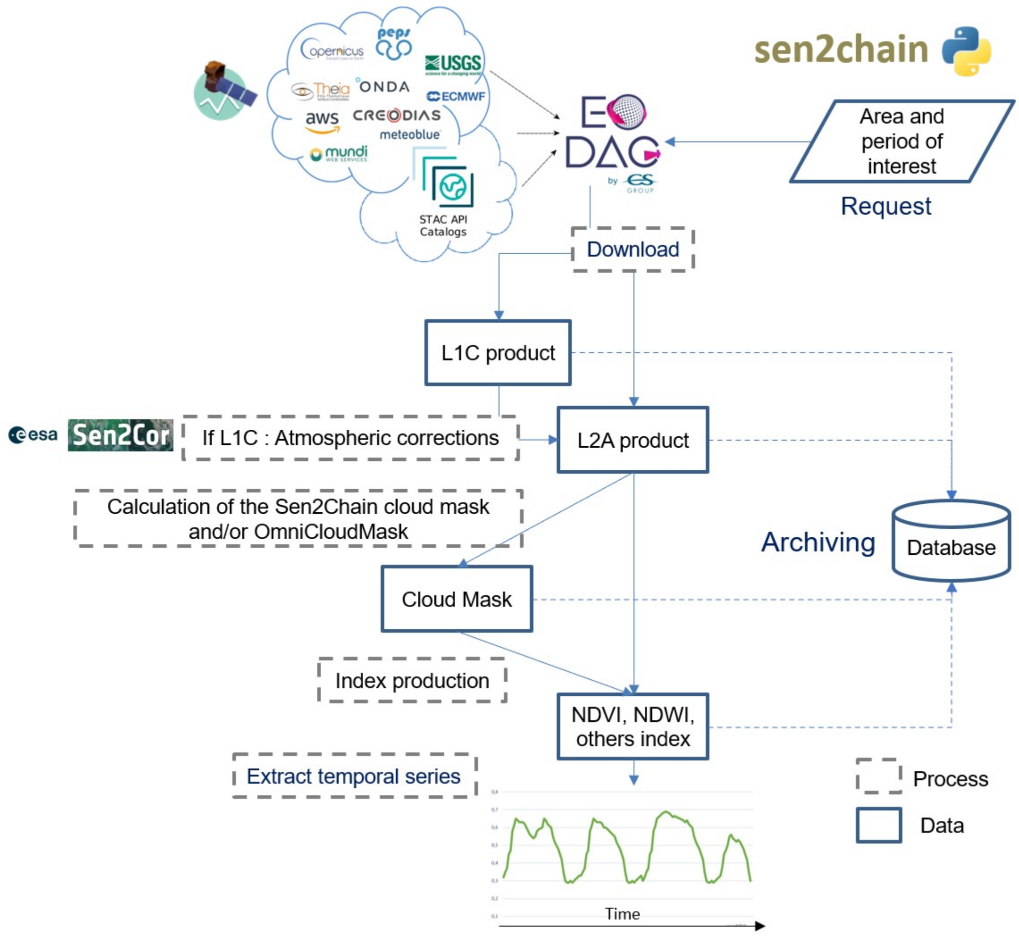

Figure 1: Key features of the Sen2Chain tool, including downloading, storing and processing Sentinel-2 images and their products

Sen2Chain uses object-oriented programming to facilitate and automate tasks. The `Tile` class handles image operations at the tile level, while the `Job` class has been implemented to facilitate the management of the image database and the execution of all the different processing tasks. Jobs enable a set of tiles to be processed as a complete workflow (from downloading to index production) in a recurring and automated manner. The `TimeSeries` class allows users to extract radiometric index data over specific sites, from point or polygon vector files, and timeframes, with customizable cloud masks. All processing steps of the chain have been designed and optimized to handle large volumes of data, in particular through parallelization using the Python modules `threading` (downloading) and `multiprocessing` (computing cloudmasks, indices, and time series extraction) when possible to fully leverage multiple processors of the computing machine, distributing the load and reducing the computation time.

Tutorials and Documentation

Sen2Chain provides comprehensive documentation including installation instructions, cookbook, and extensive details of the different classes and methods hosted on the opensource platform ReadTheDocs.

Research impact statement

The development of Sen2Chain was initiated in 2017 for research projects S2-Malaria (funded by CNES - TOSCA program, 2016-2018) and ClimHealth (accredited by the Space Climate Observatory and funded by CNES, 2020-2021), using environmental data in epidemiological surveillance systems, and ReNovRisk-Impact (funded by FEDER INTERREG V Ocean Indien 2014-2020), mapping the impacts of cyclones (Mouquet et al., 2020; Tulet et al., 2021). This tool has been under continuous development since its creation, with new improvements and features progressively added to meet research needs. It has been used in several research projects within different research units (Blanchard et al., 2024; Evans et al., 2025), and currently several operational online tools using Sen2Chain make it possible to track the evolution of health risks in real time (ClimHealth - Leptospirosis Yangon), to extract time series data on areas of interest (Sen2Extract) or to monitor the evolution of mangroves (Mangmap).

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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